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ODD END SEMESTER EXAMINATION DEC 2025

Name of the Course: B. Tech
Name of the Paper: Mechanics of Solids

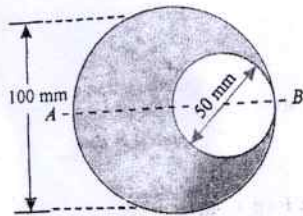
Semester: 3th
Paper Code: TCE -303

Maximum Marks: 100

Time: 3 Hours

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

		(20marks)	
Q1			
(a)	(i) What are the assumption in Torsion equation? (ii) Derive the expression for Torsion equation.		
(b)	A circular hole of 50 mm diameter is cut out from a circular disc of 100 mm diameter as shown in figure. Find the centre of gravity of the section from A.		CO1, 2 & 5
			
(c)	The principal tensile stresses at a point across two perpendicular planes are 80 MN/m ² and 40 MN/m ² . Find the normal, tangential and resultant stresses and its obliquity on a plane at 20° with the major principal plane.		
Q2		(20marks)	
(a)	Define the terms: Elasticity, Hardness, Young's modulus, Modulus of rigidity and Hooks law.		
(b)	A straight circular rod tapering from diameter D at one end to a diameter d at the other end is subjected to an axial load P. Obtain an expression for the elongation of the rod.		CO2 and CO5
(c)	Calculate the safe compressive load on a hollow cast iron column (one end rigidly fixed and the other hinged) of 150 mm external diameter, 100 mm internal diameter and 10 m length. Use Euler's formula with factor of safety of 5, and E= 95 GN/m ² .		
Q3		(20marks)	
(a)	Draw the SFD and BMD for a cantilever of length L carrying a gradually varying load from zero at the free end to w per unit length at fixed end.		
(b)	(i) What do you mean by simple bending or pure bending? What are the assumptions made in the theory of simple bending? (ii) A steel plate of width 60 mm and of thickness 10 mm is bent into a circular arc of radius 10 m. Determine the maximum stress induced and the bending moment which will produce the maximum stress. Take E = 2x 10 ⁵ N/mm ² .		CO3
(c)	A rectangular beam of 300 mm deep is simply supported over a span of 4		

	metres. Determine the uniformly distributed load per metre which the beam may carry, if the bending stress should not exceed 120 N/mm^2 . Take $I = 8 \times 10^6 \text{ mm}^4$.	
	(20marks)	
Q4		
(a)	(i) what are the assumption in bending equation? (ii) Derive the expression for the bending equation.	CO4 & 5
(b)	A cantilever 1.5 m long carries a uniformly distributed load over the entire length. Find the deflection at the free end if the slope at the free end is 1.5° .	
(c)	A column of timber section 10 cm x 15 cm is 5 m long both end being fixed. If the young's modulus for timber = 17.5 kN/mm^2 , determine (i) crippling load, and (ii) safe load for the column if factor of safety = 3.	
	(20marks)	
Q5		
(a)	What must be the length of a 5 mm diameter aluminium wire so that it can be twisted through one complete revolution without exceeding a shearing stress of 42 MN/m^2 . Take $G = 27 \text{ GN/m}^2$.	CO5
(b)	A beam 5 m long, simply supported at its ends, carries a point load W at its centre. If the slope at the ends of the beam is not to exceed 1° , find the deflection at the centre of the beam.	
(c)	A symmetrical section 200 mm deep has a moment of inertia of $2.26 \times 10^{-5} \text{ m}^4$ about its neutral axis. Determine the longest span over which, when simply supported, the beam would carry a uniformly distributed load of 4 kN/m run without the stress due to bending exceeding 125 MN/m^2 .	